PART IIA, PAPER 3, SUPERVISION 4
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SOLUTIONS TO PROBLEMS

S4.1 Give me an instrument suitable enough, and I shall move the endogeneity! Consider the following bivariate linear regression:

$$Y_i = \alpha + \beta T_i + u_i, \quad i = 1, \ldots, n, \tag{1}$$

where α and β are unknown parameters, u_i is an error term and T_i is a binary treatment regressor defined as

$$T_i = \begin{cases} 1 & \text{if individual } i \text{ is in the treated group} \\ 0 & \text{if individual } i \text{ is in the control group} \end{cases}$$

(a) Describe an empirical, real-life example where such an equation might arise.

(b) Why might u_i be heteroskedastic in your example? What are the implications of heteroskedasticity for the OLS estimator of β?

(c) Why might T_i be endogenous in your example? What are the implications of endogeneity for the OLS estimator of β?

Solution: The explanatory variable T_i is endogenous if $\mathbb{E}[u_i|T_i] \neq 0$. Endogeneity implies that $\text{Cov}(T_i, u_i) \neq 0$. The OLS estimator of β from (1), denoted by $\hat{\beta}_{\text{OLS}}$, is given by

$$\hat{\beta}_{\text{OLS}} = \frac{\widehat{\text{Cov}}(T_i, Y_i)}{\widehat{\text{Var}}(T_i)} = \frac{S_{TY}}{S_{TT}} = \frac{\sum_{i=1}^n (T_i - \bar{T}) Y_i}{\sum_{i=1}^n (T_i - \bar{T})^2}, \tag{2}$$

where $\widehat{\text{Cov}}$ and $\widehat{\text{Var}}$ denote the sample covariance and the sample variance, respectively. In this expression, substitute for Y_i from (1), to show that

$$\hat{\beta}_{\text{OLS}} = \frac{\sum_{i=1}^n (T_i - \bar{T}) Y_i}{\sum_{i=1}^n (T_i - \bar{T})^2} = \frac{\sum_{i=1}^n (T_i - \bar{T})(\alpha + \beta T_i + u_i)}{\sum_{i=1}^n (T_i - \bar{T})^2}. \tag{3}$$

Recall that this expression can simplified to

$$\hat{\beta}_{\text{OLS}} = \beta + \frac{\sum_{i=1}^n (T_i - \bar{T}) u_i}{\sum_{i=1}^n (T_i - \bar{T})^2}. \tag{4}$$

Now, since $\mathbb{E}[u_i|T_i] \neq 0$, we can show that conditional on the full vector of explanatory variables for all individuals, $\mathbf{T} = (T_1, \ldots, T_n)^\intercal$, the conditional expectation of the numerator from (4) is

$$\mathbb{E}\left[\sum_{i=1}^n (T_i - \bar{T}) u_i \,|\, \mathbf{T}\right] = \sum_{i=1}^n (T_i - \bar{T}) \mathbb{E}[u_i|T_i] \neq 0.$$

Therefore, we can conclude that the OLS estimator is not unbiased, where we use the law of iterated expectations to compute the unconditional expectation of $\hat{\beta}_{\text{OLS}}$:

$$\mathbb{E}\left[\hat{\beta}_{\text{OLS}}\right] = \mathbb{E}\left[\mathbb{E}\left[\beta_1 + \frac{\sum_{i=1}^n (T_i - \bar{T}) u_i}{\sum_{i=1}^n (T_i - \bar{T})^2} \,\middle|\, \mathbf{T}\right]\right] \neq \beta. \tag{5}$$

To analyze consistency, observe that the OLS estimator can be expressed as

$$\hat{\beta}_{\text{OLS}} = \beta + \frac{\frac{1}{n}\sum_{i=1}^{n}\left(T_i - \bar{T}\right)u_i}{\frac{1}{n}\sum_{i=1}^{n}\left(T_i - \bar{T}\right)^2}. \tag{6}$$

Use the law of large numbers to show that as $n \to \infty$,

$$\frac{1}{n}\sum_{i=1}^{n}\left(T_i - \bar{T}\right)u_i \xrightarrow{\text{p}} \mathbb{E}\left[\left(T_i - \bar{T}\right)u_i\right] = \text{Cov}\left(T_i, u_i\right) \neq 0, \tag{7}$$

$$\frac{1}{n}\sum_{i=1}^{n}\left(T_i - \bar{T}\right)^2 \xrightarrow{\text{p}} \mathbb{E}\left[\left(T_i - \bar{T}\right)^2\right] = \text{Var}\left(T_i\right) = \sigma_T^2 < \infty. \tag{8}$$

In the second expression, $\text{Var}\left(T_i\right) = \sigma_T^2 < \infty$, is an additional assumption. Thus, we can apply the Slutsky's theorem to show that since $\text{Cov}\left(T_i, u_i\right) \neq 0$, the OLS estimator $\hat{\beta}_{\text{OLS}}$ is inconsistent for β:

$$\hat{\beta}_{\text{OLS}} \xrightarrow{\text{p}} \beta + \frac{\text{Cov}\left(T_i, u_i\right)}{\text{Var}\left(T_i\right)} \neq \beta. \tag{9}$$

(d) Suppose T_i is endogenous and a single instrument Z_i is available for it. State conditions that Z_i must satisfy in order to qualify as a suitable instrument for T_i.

Solution: Two conditions must be satisfied:

(1) Validity (exclusion restriction): Z_i must affect Y_i only through its influence on T_i, and not in any other way. In other words, Z_i must be exogenous with respect to u_i in regression (1), or formally $\text{Cov}\left(Z_i, u_i\right) = 0$.

(2) Relevance (informative instrument): Z_i must have non-trivial explanatory power for T_i, namely $\text{Cov}\left(T_i, Z_i\right) \neq 0$.

(e) Show that the population slope from (1) satisfies

$$\beta = \frac{\text{Cov}\left(Z_i, Y_i\right)}{\text{Cov}\left(Z_i, T_i\right)}. \tag{10}$$

Solution: Consider the numerator in (10)

$$\text{Cov}\left(Z_i, Y_i\right) = \text{Cov}\left(Z_i, \alpha + \beta T_i + u_i\right) = \beta\text{Cov}\left(Z_i, T_i\right) + \text{Cov}\left(Z_i, u_i\right) = \beta\text{Cov}\left(Z_i, T_i\right),$$

where following the assumption of validity of the instrument, $\text{Cov}\left(Z_i, u_i\right) = 0$. Now, divide this by $\text{Cov}\left(Z_i, T_i\right)$ on both sides to obtain expression (10).

(f) Describe the Two-stage Least Squares (2SLS) procedure for estimating β. Derive the expression for the 2SLS estimator of β, denoted $\hat{\beta}_{\text{IV}}$.

Solution: In the first-stage, regress T_i on Z_i:

$$T_i = \delta_0 + \delta_1 Z_i + v_i, \quad i = 1, \ldots, n, \tag{11}$$

where v_i are the error terms. We assume that Z_i is exogenous with respect to v_i, that is $\mathbb{E}\left[v_i | Z_i\right] = 0$, and that Z_i is informative for T_i, that is $\text{Cov}\left(Z_i, T_i\right) \neq 0$. Then, the OLS estimator of δ_1 from (11) is given by

$$\hat{\delta}_1 = \frac{\widehat{\text{Cov}}\left(Z_i, T_i\right)}{\widehat{\text{Var}}\left(Z_i\right)} = \frac{S_{ZT}}{S_{ZZ}}. \tag{12}$$

Note that $\widehat{\text{Cov}}\left(Z_i, T_i\right) \neq 0$ is required for the instrument to have predictive power for T_i. The corresponding predicted values of T_i from the first-stage regression is then computed using $\hat{T}_i = \hat{\delta}_0 + \hat{\delta}_1 Z_i$, for all $i = 1, \ldots, n$. Now, in the second-stage, estimate the following regression:

$$Y_i = \alpha + \beta \hat{T}_i + u_i, \quad i = 1, \ldots, n. \tag{13}$$

The OLS estimator of β from regression (13), which we denote by $\hat{\beta}_{\text{IV}}$, to distinguish it from the OLS estimator $\hat{\beta}_{\text{OLS}}$, which is the estimator obtained by estimating β from regression (1) directly, is given by

$$\hat{\beta}_{\text{IV}} = \frac{\widehat{\text{Cov}}\left(\hat{T}_i, Y_i\right)}{\widehat{\text{Var}}\left(\hat{T}_i\right)} = \frac{\sum_{i=1}^{n}\left(\hat{T}_i - \bar{\hat{T}}\right) Y_i}{\sum_{i=1}^{n}\left(\hat{T}_i - \bar{\hat{T}}\right)^2}. \tag{14}$$

Consider the numerator and denominator separately

$$\begin{aligned}
\widehat{\text{Cov}}\left(\hat{T}_i, Y_i\right) &= \widehat{\text{Cov}}\left(\hat{\delta}_0 + \hat{\delta}_1 Z_i, Y_i\right) = \hat{\delta}_1 \widehat{\text{Cov}}\left(Z_i, Y_i\right), \\
\widehat{\text{Var}}\left(\hat{T}_i\right) &= \widehat{\text{Var}}\left(\hat{\delta}_0 + \hat{\delta}_1 Z_i\right) = \hat{\delta}_1^2 \widehat{\text{Var}}\left(Z_i\right).
\end{aligned}$$

Putting these two together

$$\hat{\beta}_{\text{IV}} = \frac{\widehat{\text{Cov}}\left(\hat{T}_i, Y_i\right)}{\widehat{\text{Var}}\left(\hat{T}_i\right)} = \frac{\hat{\delta}_1 \widehat{\text{Cov}}\left(Z_i, Y_i\right)}{\hat{\delta}_1^2 \widehat{\text{Var}}\left(Z_i\right)} = \left(\frac{1}{\hat{\delta}_1}\right)\left(\frac{\widehat{\text{Cov}}\left(Z_i, Y_i\right)}{\widehat{\text{Var}}\left(Z_i\right)}\right).$$

Recall the expression for $\hat{\delta}_1$ from (12):

$$\hat{\beta}_{\text{IV}} = \left(\frac{\widehat{\text{Var}}\left(Z_i\right)}{\widehat{\text{Cov}}\left(Z_i, T_i\right)}\right)\left(\frac{\widehat{\text{Cov}}\left(Z_i, Y_i\right)}{\widehat{\text{Var}}\left(Z_i\right)}\right) = \frac{\widehat{\text{Cov}}\left(Z_i, Y_i\right)}{\widehat{\text{Cov}}\left(Z_i, T_i\right)}. \tag{15}$$

The IV estimator is given by the ratio of the sample covariance of the instrument with the dependent variable and its covariance with the endogenous explanatory variable. We can therefore write

$$\hat{\beta}_{\text{IV}} = \frac{S_{ZY}}{S_{ZT}} = \frac{\sum_{i=1}^{n}\left(Z_i - \bar{Z}\right) Y_i}{\sum_{i=1}^{n}\left(Z_i - \bar{Z}\right) T_i}. \tag{16}$$

Finally, observe that we can divide the numerator and the denominator by the sample covariance of the instrument, S_{ZZ}, to obtain:

$$\hat{\beta}_{\text{IV}} = \left(\frac{S_{ZY}}{S_{ZT}}\right)\left(\frac{S_{ZZ}}{S_{ZZ}}\right) = \left(\frac{S_{ZY}}{S_{ZZ}}\right) / \left(\frac{S_{ZT}}{S_{ZZ}}\right) = \frac{\hat{\gamma}_1}{\hat{\delta}_1}. \tag{17}$$

Note that $\hat{\gamma}_1$ is the OLS estimator of the slope coefficient from the regression of Y_i on Z_i, and $\hat{\delta}_1$ is the OLS estimator of the slope coefficient from the regression of T_i on Z_i, namely

$$Y_i = \gamma_0 + \gamma_1 Z_i + \varepsilon_i \implies \hat{\gamma}_1 = S_{ZY}/S_{ZZ} \tag{18}$$

$$T_i = \delta_0 + \delta_1 Z_i + v_i \implies \hat{\delta}_1 = S_{ZT}/S_{ZZ} \tag{19}$$

Moreover, it can be shown that the 2SLS estimator is consistent, $\hat{\beta}_{\text{IV}} \xrightarrow{\text{p}} \beta$, if the two assumptions hold: $\text{Cov}(Z_i, u_i) = 0$ and $\text{Cov}(Z_i, T_i) \neq 0$.

(g) Suppose that the instrument Z_i is a binary variable. Show that in this case, the 2SLS estimator for β you derived in the previous part can be written as

$$\hat{\beta}_{\text{IV}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{T}_1 - \bar{T}_0}, \tag{20}$$

where $\bar{Y}_0$ and $\bar{T}_0$ are the sample averages of Y_i and T_i over the part of the sample where $Z_i = 0$, and $\bar{Y}_1$ and $\bar{T}_1$ are the sample averages of Y_i and T_i over the part of the sample where $Z_i = 1$. In this case, when the instrument is a binary variable, the IV estimator is called the Wald estimator.

Solution: Compare the IV estimator in expression (16) we derived in the previous part with that in expression (20). It is sufficient to show that the numerator part holds, and the denominator will follow by the same logic. Hence, we want to show that $\hat{\gamma}_1 = S_{ZY}/S_{ZZ} = \bar{Y}_1 - \bar{Y}_0$. We can do that using three methods, pick the one that you like the most.

Method 1: Quick and cheeky. Start with the numerator from (16), and we want to show that

$$\sum_{i=1}^{n} \left(Z_i - \bar{Z}\right) Y_i = \bar{Y}_1 - \bar{Y}_0.$$

The sub-sample averages are given by

$$\bar{Y}_1 = \sum_{i:Z_i=1}^{n} Y_i, \qquad \bar{Y}_0 = \sum_{i:Z_i=0}^{n} Y_i.$$

Note that we are showing that if Z_i is a binary variable, then the OLS estimator of the slope coefficient from the regression $Y_i = \gamma_0 + \gamma_1 Z_i + \varepsilon_i$, is given by $\hat{\gamma}_1 = \bar{Y}_1 - \bar{Y}_0$. Consider the population regression and the expected values of Y_i, conditional on the two possible values of the explanatory variable Z_i:

$$\begin{aligned} \mathbb{E}\left[Y_i | Z_i = 0\right] &= \gamma_0, \\ \mathbb{E}\left[Y_i | Z_i = 1\right] &= \gamma_0 + \gamma_1. \end{aligned}$$

Therefore, the population slope is given by $\gamma_1 = \mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i | Z_i = 0\right]$. The estimator is given by the corresponding sample analogue of the difference in the conditional expectations of Y_i, which is of course simply $\hat{\gamma}_1 = \bar{Y}_1 - \bar{Y}_0$. This is the difference between the sample averages of Y_i where $Z_i = 1$ and $Z_i = 0$, respectively. Similarly, we can show that the OLS estimator from the regression $T_i = \delta_0 + \delta_1 Z_i + v_i$ is given by $\hat{\delta}_1 = \bar{T}_1 - \bar{T}_0$. The expression in (20) follows.

Method 2: Law of iterated expectations. The alternative method uses the law of iterated expectations, so is an excellent exercise in its own right. We want to consider the expression for the population coefficient and analyze the numerator and

the denominator separately by expressing covariances as expectations from the first principles,

$$\beta = \frac{\mathrm{Cov}\left(Z_i, Y_i\right)}{\mathrm{Cov}\left(Z_i, T_i\right)}.$$

Consider the numerator from this expression:

$$\mathrm{Cov}\left(Z_i, Y_i\right) = \mathbb{E}\left[Z_i Y_i\right] - \mathbb{E}\left[Z_i\right]\mathbb{E}\left[Y_i\right].$$

Since Z_i only takes two possible values, it is easy to see that $\mathbb{P}\left(Z_i = 1\right) = \mathbb{E}\left[Z_i\right]$. Therefore, $\mathbb{P}\left(Z_i = 0\right) = 1 - \mathbb{E}\left[Z_i\right]$. We can use these probability expressions to compute $\mathbb{E}\left[Z_i Y_i\right]$ using the law of iterated expectations by conditioning it on Z_i, namely

$$\begin{aligned}
\mathbb{E}\left[Z_i Y_i\right] &= \mathbb{E}\left[\mathbb{E}\left[Z_i Y_i | Z_i\right]\right] \\
&= \mathbb{E}\left[\left(0 \times Y_i | Z_i = 0\right)\mathbb{P}\left(Z_i = 0\right) + \left(1 \times Y_i | Z_i = 1\right)\mathbb{P}\left(Z_i = 1\right)\right] \\
&= \mathbb{E}\left[Y_i | Z_i = 1\right]\mathbb{E}\left[Z_i\right].
\end{aligned}$$

Do the same for $\mathbb{E}\left[Y_i\right]$:

$$\begin{aligned}
\mathbb{E}\left[Y_i\right] &= \mathbb{E}\left[\mathbb{E}\left[Y_i | Z_i\right]\right] \\
&= \mathbb{E}\left[\left(Y_i | Z_i = 0\right)\mathbb{P}\left(Z_i = 0\right) + \left(Y_i | Z_i = 1\right)\mathbb{P}\left(Z_i = 1\right)\right] \\
&= \mathbb{E}\left[\left(Y_i | Z_i = 0\right)\left(1 - \mathbb{E}\left[Z_i\right]\right) + \left(Y_i | Z_i = 1\right)\mathbb{E}\left[Z_i\right]\right] \\
&= \mathbb{E}\left[Y_i | Z_i = 0\right]\left(1 - \mathbb{E}\left[Z_i\right]\right) + \mathbb{E}\left[Y_i | Z_i = 1\right]\mathbb{E}\left[Z_i\right].
\end{aligned}$$

Use these expressions for $\mathbb{E}\left[Z_i Y_i\right]$ and $\mathbb{E}\left[Y_i\right]$ to simplify covariance:

$$\begin{aligned}
\mathrm{Cov}\left(Z_i, Y_i\right) &= \mathbb{E}\left[Z_i Y_i\right] - \mathbb{E}\left[Z_i\right]\mathbb{E}\left[Y_i\right] \\
&= \mathbb{E}\left[Y_i | Z_i = 1\right]\mathbb{E}\left[Z_i\right] - \mathbb{E}\left[Z_i\right]\mathbb{E}\left[Y_i\right] \\
&= \mathbb{E}\left[Z_i\right]\Big(\mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i\right]\Big) \\
&= \mathbb{E}\left[Z_i\right]\Big(\mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i | Z_i = 0\right]\left(1 - \mathbb{E}\left[Z_i\right]\right) - \mathbb{E}\left[Y_i | Z_i = 1\right]\mathbb{E}\left[Z_i\right]\Big) \\
&= \mathbb{E}\left[Z_i\right]\left(1 - \mathbb{E}\left[Z_i\right]\right)\Big(\mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i | Z_i = 0\right]\Big).
\end{aligned}$$

Using the same method, we can also show that

$$\mathrm{Cov}\left(Z_i, T_i\right) = \mathbb{E}\left[Z_i\right]\left(1 - \mathbb{E}\left[Z_i\right]\right)\Big(\mathbb{E}\left[T_i | Z_i = 1\right] - \mathbb{E}\left[T_i | Z_i = 0\right]\Big).$$

Combine these two results, to obtain

$$\begin{aligned}
\beta &= \frac{\mathrm{Cov}\left(Z_i, Y_i\right)}{\mathrm{Cov}\left(Z_i, T_i\right)} \\
&= \frac{\mathbb{E}\left[Z_i\right]\left(1 - \mathbb{E}\left[Z_i\right]\right)\left(\mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i | Z_i = 0\right]\right)}{\mathbb{E}\left[Z_i\right]\left(1 - \mathbb{E}\left[Z_i\right]\right)\left(\mathbb{E}\left[T_i | Z_i = 1\right] - \mathbb{E}\left[T_i | Z_i = 0\right]\right)} \\
&= \frac{\mathbb{E}\left[Y_i | Z_i = 1\right] - \mathbb{E}\left[Y_i | Z_i = 0\right]}{\mathbb{E}\left[T_i | Z_i = 1\right] - \mathbb{E}\left[T_i | Z_i = 0\right]}
\end{aligned}$$

Finally, replace the population moments with the sample counterparts:

$$\hat{\beta}_{\mathrm{IV}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{T}_1 - \bar{T}_0}.$$

Method 3. Brute force derivation. We can also show the same thing using a direct method. We will only show the result for the numerator, namely that in regression (18), $\hat{\gamma}_1 = \bar{Y}_1 - \bar{Y}_0$. Denote the number of individuals for whom $Z_i = 1$ by n_1, and the number of individuals for whom $Z_i = 0$ by n_0, such that $n_0 + n_1 = n$ is the total number of observations. Rank the individuals so that the first n_1 of them are such that their $Z_i = 1$, and the remaining n_0 of them are such that $Z_i = 0$. As a result, the mean of Z_i can be written simply as

$$\bar{Z} = \frac{1}{n}\sum_{i=1}^{n} Z_i = \frac{n_1}{n}. \tag{21}$$

Observe that $\bar{Z}$ is the proportion of individuals for whom $Z_i = 1$. Also,

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i, \qquad \bar{Y}_1 = \frac{1}{n_1}\sum_{i=1}^{n_1} Y_i, \qquad \bar{Y}_0 = \frac{1}{n_0}\sum_{i=n_1+1}^{n} Y_i,$$

where $\bar{Y}$ is the overall mean of Y_i, $\bar{Y}_1$ is the mean of Y_i for those observations where $Z_i = 1$, and $\bar{Y}_0$ is the mean of Y_i for those observations with $Z_i = 0$. Recall that the OLS estimator of the slope coefficient of a regression of Y_i on Z_i is given by

$$\begin{aligned}\hat{\gamma}_1 &= \frac{\widehat{\mathrm{Cov}}\left(Y_i, Z_i\right)}{\widehat{\mathrm{Var}}\left(Z_i\right)} = \frac{\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right) Z_i}{\sum_{i=1}^{n}\left(Z_i - \bar{Z}\right)^2} \\ &= \frac{\sum_{i=1}^{n} Y_i Z_i - \sum_{i=1}^{n} \bar{Y} Z_i}{\sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2} = \frac{\sum_{i=1}^{n} Y_i Z_i - n\bar{Y}\bar{Z}}{\sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2}.\end{aligned}$$

Consider the denominator of this expression:

$$\sum_{i=1}^{n} Z_i^2 - n\bar{Z}^2 = n\bar{Z} - n\bar{Z}^2 = n\bar{Z}\left(1 - \bar{Z}\right) = \frac{n_0 n_1}{n}.$$

Here we used the fact that $Z_i^2 = Z_i$ to simplify $\sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n} Z_i = nZ$, and consequently that $\left(1 - \bar{Z}\right) = n_0/n$. In the numerator note that the first term can be simplified as

$$\sum_{i=1}^{n} Y_i Z_i = \sum_{i=1}^{n} Y_i \mathbf{1}\left\{Z_i = 1\right\} = \sum_{i=1}^{n_1} Y_i = n_1 \bar{Y}_1,$$

where $\mathbf{1}\{\cdot\}$ is the indicator function such that

$$\mathbf{1}\left\{Z_i = 1\right\} = \begin{cases} 1 & \text{if } Z_i = 1 \\ 0 & \text{if } Z_i = 0 \end{cases}$$

The second term in the numerator is given by

$$n\bar{Y}\bar{Z} = n\bar{Y}\left(\frac{n_1}{n}\right) = n_1\bar{Y}.$$

The OLS estimator then reduces to

$$\hat{\gamma}_1 = n\left(\frac{n_1\left(\bar{Y}_1 - \bar{Y}\right)}{n_0 n_1}\right) = \frac{n}{n_0}\left(\bar{Y}_1 - \bar{Y}\right). \tag{22}$$

We can use the expressions for the average incomes to simplify the above

$$\begin{aligned}
\hat{\gamma}_1 &= \frac{n}{n_0}\left[\frac{1}{n_1}\sum_{i=1}^{n_1} Y_i - \frac{1}{n}\sum_{i=1}^{n} Y_i\right] \\
&= \frac{n}{n_0}\left[\frac{1}{n_1}\sum_{i=1}^{n_1} Y_i - \frac{1}{n}\left(\sum_{i=1}^{n_1} Y_i + \sum_{i=n_1+1}^{n} Y_i\right)\right] \\
&= \frac{n}{n_0}\left[\left(\frac{1}{n_1}\sum_{i=1}^{n_1} Y_i - \frac{1}{n}\sum_{i=1}^{n_1} Y_i\right) + \sum_{i=n_1+1}^{n} Y_i\right] \\
&= \frac{n}{n_0}\left[\frac{n-n_1}{n_1 n}\sum_{i=1}^{n_1} Y_i - \frac{1}{n}\sum_{i=n_1+1}^{n} Y_i\right] \\
&= \frac{n}{n_0}\left[\frac{n_0}{n_1 n}\sum_{i=1}^{n_1} Y_i - \frac{1}{n}\sum_{i=n_1+1}^{n} Y_i\right] \\
&= \frac{1}{n_1}\sum_{i=1}^{n_1} Y_i - \frac{1}{n_0}\sum_{i=n_1+1}^{n} Y_i \\
&= \bar{Y}_1 - \bar{Y}_0.
\end{aligned}$$

Using the same method we can show that in regression (19), $\hat{\delta}_1 = \bar{T}_1 - \bar{T}_0$, and the result follows.

(h) Suggest a suitable instrument for your example. Justify your choice.

S4.2 Moving closer to a higher wage. Literally.

Consider the following linear model of log wages (`lwage`) explained using years of schooling (`sch`), years of experience and its square (`exp`, `exp`^{2}), and 3 dummy variables indicating whether the individual was black (`black`), lived in the south (`south`), and lived in a metropolitan area (`met`):

$$\texttt{lwage}_i = \beta_0 + \beta_1\texttt{sch}_i + \beta_2\texttt{exp}_i + \beta_3\texttt{exp}_i^2 + \beta_4\texttt{black}_i + \beta_5\texttt{south}_i + \beta_6\texttt{met}_i + \varepsilon_i, \quad (23)$$

where ε_i is an unobserved error term, and $i = 1, \ldots, n$ indexes individuals. A reduced form model for schooling is written as

$$\texttt{sch}_i = \alpha + \mathbf{Z}_i^\intercal \boldsymbol{\pi} + v_i, \quad (24)$$

where $\mathbf{Z}_i$ is 6×1 vector of explanatory variables including the instrument $\texttt{nearcol}_i$, which is a binary variable equal to one if the individual lived near college, and zero otherwise, and all the exogenous variables in (23), $\boldsymbol{\pi}$ is a 6×1 vector of unknown parameters, α is an unknown scalar parameter and v_i is an error term.

$$\mathbf{Z}_i = \begin{bmatrix} \texttt{nearcol}_i \\ \texttt{exp}_i \\ \texttt{exp}_i^2 \\ \texttt{black}_i \\ \texttt{south}_i \\ \texttt{met}_i \end{bmatrix}, \qquad \boldsymbol{\pi} = \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \\ \pi_4 \\ \pi_5 \\ \pi_6 \end{bmatrix}.$$

(a) Why might $\text{Cov}(v_i, \varepsilon_i)$ be non-zero?

Solution: This is the case when v_i captures some unobserved factors that influence education, such as intellect, motivation, which also appear in ε_i, that is they affect wage, controlling for demographics, schooling and experience.

(b) Assuming that $\texttt{sch}_i$ is endogenous in (23). How might you use the reduced form equation for $\texttt{sch}_i$ in (24) in conjunction with (23) to identify the parameter β_1?

Table 1: Wage equation estimated by OLS and IV

	OLS `lwage`	IV `lwage`
`sch`	0.074 (21.11)	0.133 (2.58)
`exp`	0.084 (12.58)	0.056 (2.15)
$\texttt{exp}^2$	−0.002 (−7.05)	−0.008 (−0.06)
`black`	−0.189 (−10.76)	−0.103 (−1.33)
`south`	−0.125 (−8.26)	−0.098 (−3.41)
`met`	0.181 (10.37)	0.108 (2.17)
`const`	4.734 (70.02)	4.066 (6.68)
R^2	0.291	
F-stat	204.93	

t-statistics in parenthesis

Solution: Use the two-stage least square (2SLS). The reduced-form equation for $\texttt{sch}_i$ is the first-stage, where we extract the exogenous part of $\texttt{sch}_i$ using the instrument "lived near college" (and control variables from the original log wage regression). Use the predicted values from the first-stage regression, $\widehat{\texttt{sch}}_i$, as instrument for $\texttt{sch}_i$ in the original regression (23):

$$\texttt{lwage}_i = \gamma_0 + \gamma_1 \widehat{\texttt{sch}}_i + \gamma_2 \texttt{exp}_i + \gamma_3 \texttt{exp}_i^2 + \gamma_4 \texttt{black}_i + \gamma_5 \texttt{south}_i + \gamma_6 \texttt{met}_i + e_i$$

The OLS estimator $\hat{\gamma}_1$ from the second-stage regression is the 2SLS estimator of β_1, which is consistent and asymptotically normal, provided the instrument is exogenous and relevant.

(c) Table 1 reports results from estimating two models from (23). The first uses direct OLS estimation, and second uses 2SLS or IV estimation. In Table 1, t-statistics for the corresponding coefficients are given in parenthesis under the coefficient estimates.

The instrument $\texttt{nearcol}_i$ used for $\texttt{sch}_i$ in the first-stage regression of IV estimation. R^2 for the reduced-form regression for $\texttt{sch}_i$ is 0.119. Data is a sample of 3010 men taken from the US National Longitudinal Survey of Young Men in 1976. Provide a careful comparison of the results based on the OLS and IV estimators. Discuss the choice of the instrument.

Solution: First, note that $\hat{\beta}_1^{\text{2SLS}}$ is almost twice as large as $\hat{\beta}_1^{\text{OLS}}$,so there is a negative bias in OLS. This could potentially be due to a random measurement error in education, resulting in an attenuation bias (after all, these data are self-reported). Also, the model certainly suffers from omitted variable bias: relevant controls are missing such as socioeconomic and parental background, personality traits such as motivation and perseverance. Looking at much lower t-statistics of the coefficient we can infer that the standard errors of the estimated coefficients are much higher for 2SLS than for OLS, as expected. However, $\texttt{sch}_i$ is still significant in the 2SLS model.

Exogeneity of instrumental variable $\texttt{nearcol}_i$ could potentially be violated for a variety of reason – for example if wealthy parents choose to live near college. This would mean that $\texttt{nearcol}_i$ is correlated with unobserved factors that also affect wage. We are not presented with the estimation results from the first stage, so it's impossible to judge whether the instrument is relevant.

S4.3 "Get born first, or die tryin'...": the pecking order of returns to education. The data for this question is in datafile `wage2.dta`. Suppose we want to estimate the returns to education ($\texttt{educ}_i$) using the following bivariate regression model for log wages ($\texttt{lwage}_i$)

$$\texttt{lwage}_i = \beta_0 + \beta_1 \texttt{educ}_i + u_i, \quad i = 1, \ldots, n. \tag{25}$$

(a) Explain why $\texttt{educ}_i$ is likely to be endogenous in regression (25). How will endogeneity of $\texttt{educ}_i$ affect the OLS estimators of the returns to education?

Solution: Education is endogenous, namely $\mathbb{E}\left[u_i|\texttt{educ}_i\right] \neq 0$, because the unobserved component u_i contains (omitted) variables that affect wages and are correlated with education, such as for example parental income and own ability or motivation. Endogeneity of education makes the OLS estimator of β_1 biased and inconsistent.

(b) In an attempt to correct for endogeneity of $\texttt{educ}_i$, it is proposed to use family variables such as the number of siblings, $\texttt{sibs}_i$, as an instrument for $\texttt{educ}_i$. Using the data provided in `wage2.dta`, the IV estimate for the returns to education is 0.122. To convince yourself that using $\texttt{sibs}_i$ as an IV for $\texttt{educ}_i$ is not the same as just plugging $\texttt{sibs}_i$ in for $\texttt{educ}_i$ and running an OLS regression, run the regression of $\texttt{lwage}_i$ on $\texttt{sibs}_i$ and explain your findings.

Solution: For $\texttt{sibs}_i$ to be a reasonable instrument for $\texttt{educ}_i$, it must be the case that $\texttt{sibs}_i$ is (i) randomly assigned, which to some extent is true, (ii) valid, that is $\mathbb{E}\left[u_i|\texttt{sibs}_i\right] = 0$, which we have to assume, and (iii) relevant $\text{Cov}\left(\texttt{sibs}_i, \texttt{educ}_i\right) \neq 0$. The first specification that we want to estimate is the regression of log wages on the numbers of siblings:

$$\texttt{lwage}_i = \gamma_0 + \gamma_1 \texttt{sibs}_i + e_i. \tag{26}$$

The corresponding OLS estimator of returns to education, or the population coefficient

β_1 from regression 25, using this direct method is given by

$$\hat{\beta}_1^{\text{OLS}} = \hat{\gamma}_1 = \frac{\widehat{\text{Cov}}\left(\texttt{sibs}_i, \texttt{lwage}_i\right)}{\widehat{\text{Var}}\left(\texttt{sibs}_i\right)}. \tag{27}$$

STATA provides us with an OLS estimate of $\hat{\beta}_1^{\text{OLS}} = -0.0279$. The second specification for estimating returns to education uses the 2SLS (IV) method. The relevance of $\texttt{sibs}_i$ as an instrument for $\texttt{educ}_i$ can be confirm by testing significance of the slope coefficient in the first-stage regression of the 2SLS estimation procedure, which includes the following regressions:

$$\texttt{educ}_i = \delta_0 + \delta_1 \texttt{sibs}_i + v_i, \tag{28}$$

$$\texttt{lwage}_i = \beta_0 + \beta_1 \widehat{\texttt{educ}}_i + u_i. \tag{29}$$

In the second stage, the predicted values for education are used. These are given by the OLS fitted values, $\widehat{\texttt{educ}}_i = \hat{\delta}_0 + \hat{\delta}_1 \texttt{sibs}_i$, where the OLS estimator of δ_1 is

$$\hat{\delta}_1 = \frac{\widehat{\text{Cov}}\left(\texttt{sibs}_i, \texttt{educ}_i\right)}{\widehat{\text{Var}}\left(\texttt{sibs}_i\right)}. \tag{30}$$

STATA provides us with an estimate of the first-stage regression slope of $\hat{\delta}_1 = -0.2279$. The corresponding t-statistic, $t_{\hat{\delta}_1} = -7.53$, suggests that it is significance. We can confirm that $\texttt{sibs}_i$ is a relevant instrument for $\texttt{educ}_i$. The corresponding 2SLS estimator of β_1 is then given by

$$\hat{\beta}_1^{\text{IV}} = \frac{\widehat{\text{Cov}}\left(\texttt{sibs}_i, \texttt{lwage}_i\right)}{\widehat{\text{Var}}\left(\texttt{sibs}_i, \texttt{educ}_i\right)} = \left(\frac{\widehat{\text{Var}}\left(\texttt{sibs}_i\right)}{\widehat{\text{Var}}\left(\texttt{sibs}_i, \texttt{educ}_i\right)}\right)\left(\frac{\widehat{\text{Cov}}\left(\texttt{sibs}_i, \texttt{lwage}_i\right)}{\widehat{\text{Var}}\left(\texttt{sibs}_i\right)}\right) = \frac{\hat{\beta}_1^{\text{OLS}}}{\hat{\delta}_1}$$

This expression was derived in the previous question, and the in last line we used expressions from (27) and 30. It is then clear that $\hat{\beta}_1^{\text{OLS}} \neq \hat{\beta}_1^{\text{IV}}$. Using the corresponding STATA estimates, we can compute

$$\hat{\beta}_1^{\text{IV}} = \hat{\beta}_1^{\text{OLS}} / \hat{\delta}_1 = -0.0279 / \left(-0.2279\right) = 0.1224.$$

But this is exactly the same estimate as the one that STATA provides when running IV estimation directly. The STATA corresponding estimation results are provided in Table 2, column (1) for OLS of wage on siblings, column (2) for the first-stage of the IV and (3) for the second stage of the IV estimation. These results are obtained using the following code.

```
>> reg lwage sibs
>> reg educ sibs
>> ivreg lwage (educ = sibs)
```

We can see that the sign of the estimated coefficient changed and is highly significant in both cases. This is because in (1), we are not measuring the relationship between education and log wages (which we expect to be positive), but rather the relationship between the number of siblings and log wages (which we can expect to be negative). When we use $\texttt{sibs}_i$ as an IV for education, we are estimating the coefficient for

education, but we get rid of the endogenous part of education (which biases the coefficient), and use only the exogenous part of its variation (which will be the part of education that is explained by the number of siblings and is not correlated with the error term u_i).

Table 2: OLS and IV estimation of returns to education

	(1) OLS `lwage`	(2) OLS `educ`	(3) IV `lwage`	(4) OLS `educ`	(5) IV `lwage`	(6) OLS `educ`	(7) IV `lwage`
`educ`			0.122		0.131		0.136
			(4.65)		(4.08)		(1.83)
`sibs`	−0.028	−0.228				−0.152	0.002
	(−4.72)	(−7.53)				(−3.83)	(0.12)
`brthord`				−0.283		−0.152	
				(−6.11)		(−2.67)	
`const`	6.861	14.139	5.130	14.149	5.030	14.297	4.939
	(310.77)	(124.97)	(14.44)	(109.96)	(11.62)	(107.26)	(4.68)
n	935	935	935	852	852	852	852
R^2	0.0234	0.0573		0.0420		0.0583	

t-statistic in parenthesis

(c) The variable $\texttt{brthord}_i$ denotes the birth order: it is equal to one for a first-born child, two for a second-born child, and so on. Explain why $\texttt{educ}_i$ and $\texttt{brthord}_i$ might be negatively correlated. Regress $\texttt{educ}_i$ on $\texttt{brthord}_i$ and test whether the negative link between them is statistically significant. Use $\texttt{brthord}_i$ as an instrument for $\texttt{educ}_i$ and estimate regression (25) using 2SLS. Report and interpret your results.

Solution: It could be that families hit budget constraints and might not be able to afford as much education for children born later. We estimate returns to education using 2SLS where $\texttt{brthord}_i$ is used as an instrument for $\texttt{educ}_i$

$$\texttt{educ}_i = \delta_0 + \delta_1 \texttt{brthord}_i + v_i, \tag{31}$$

$$\texttt{lwage}_i = \beta_0 + \beta_1 \widehat{\texttt{educ}}_i + u_i. \tag{32}$$

In the second stage, the predicted values for education are given $\widehat{\texttt{educ}}_i = \hat{\delta}_0 + \hat{\delta}_1 \texttt{brthord}_i$. Use the following `STATA` code

```
>> reg educ brthord
>> ivreg lwage (educ = brthord)
```

Estimation results are reported in Table 2. Column (4) reports estimates for the first stage regression and column (5) for the second stage. Note that the IV estimate of returns to education from (5), $\hat{\beta}_1^{\text{IV}} = 0.130$, is higher than the IV estimate from (3) when we used the number of sibling as instrumental variable, $\hat{\beta}_1^{\text{IV}} = 0.122$. They are still relatively close and both significant. Also note that because of the missing data on

$\texttt{brthord}_i$, we are using fewer observation than in part (a). Different instruments might be capturing the variation in education for different subgroups of the population. This should be considered when comparing different IV estimates.

(d) Now, consider a linear regression where the number of siblings is included as a regressor in the wage equation to control for family background:

$$\texttt{lwage}_i = \beta_0 + \beta_1 \texttt{educ}_i + \beta_2 \texttt{sibs}_i + u_i. \tag{33}$$

Suppose that we want to use $\texttt{brthord}_i$ as an IV for $\texttt{educ}_i$, assuming that $\texttt{sibs}_i$ is exogenous. The reduced form of the first-stage regression for $\texttt{educ}_i$ is given by

$$\texttt{educ}_i = \pi_0 + \pi_1 \texttt{sibs}_i + \pi_2 \texttt{brthord}_i + v_i. \tag{34}$$

State and test the identification assumption.

Solution: First, observe that the predicted values for education from the first stage are given by

$$\widehat{\texttt{educ}}_i = \hat{\pi}_0 + \hat{\pi}_1 \texttt{sibs}_i + \hat{\pi}_2 \texttt{brthord}_i. \tag{35}$$

The corresponding second stage regression can be expressed as

$$\begin{aligned}
\texttt{lwage}_i &= \beta_0 + \beta_1 \widehat{\texttt{educ}}_i + \beta_2 \texttt{sibs}_i + u_i \\
&= \beta_0 + \beta_1 \left(\hat{\pi}_0 + \hat{\pi}_1 \texttt{sibs}_i + \hat{\pi}_2 \texttt{brthord}_i\right) + \beta_2 \texttt{sibs}_i + u_i \\
&= \left(\beta_0 + \beta_1 \hat{\pi}_0\right) + \left(\beta_2 + \beta_1 \hat{\pi}_1\right) \texttt{sibs}_i + \beta_1 \hat{\pi}_2 \texttt{brthord}_i + u_i
\end{aligned}$$

In this regression the coefficients from the first-stage regression, namely $\hat{\pi}_0$, $\hat{\pi}_1$ and $\hat{\pi}_2$, have been estimated using OLS and so are known at this point. Therefore, the coefficients that we do not know yet and want to identify (obtain unique estimates of) are β_0, β_1 and β_2. Note that the corresponding reduced-form (estimated) regression has the form

$$\texttt{lwage}_i = \gamma_0 + \gamma_1 \texttt{sibs}_i + \gamma_2 \texttt{brthord}_i + u_i \tag{36}$$

and the corresponding OLS estimates are $\hat{\gamma}_0$, $\hat{\gamma}_1$ and $\hat{\gamma}_2$. Identification then amounts to obtaining OLS estimates or equivalently solving for the values of $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\beta}_2$ using the known coefficient estimates from the first-stage, namely $\hat{\pi}_0$, $\hat{\pi}_1$ and $\hat{\pi}_2$, and the known coefficients from the second stage, namely $\hat{\gamma}_0$, $\hat{\gamma}_1$ and $\hat{\gamma}_2$. We then have to solve the following system of three equations for three unknown $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\beta}_2$:

$$\hat{\gamma}_0 = \hat{\beta}_0 + \hat{\beta}_1 \hat{\pi}_0, \tag{37}$$

$$\hat{\gamma}_1 = \hat{\beta}_2 + \hat{\beta}_1 \hat{\pi}_1, \tag{38}$$

$$\hat{\gamma}_2 = \hat{\beta}_1 \hat{\pi}_2. \tag{39}$$

Since we are primarily interested in identifying β_1, we can infer that its IV estimate is given by

$$\hat{\beta}_1 = \frac{\hat{\pi}_2}{\hat{\gamma}_2}. \tag{40}$$

Clearly, in order for β_1 to be identified, the exclusion restriction must be satisfied in the first-stage regression, namely $\pi_2 \neq 0$. This can be tested using a t-test for $H_0 \colon \pi_2 = 0$ in regression (33). The `STATA` estimation output for the firs-stage regression is reported in column (6) of Table 2, where we run

```
>> reg educ sibs brthord
```

The estimate $\hat{\pi}_2 = -0.1529$ and the corresponding test statistic $t_{\hat{\pi}_2} = -2.67$, which rejects the null hypothesis quite strongly. Therefore, the identification assumption seems to hold. Remember that we need partial correlation, that is the instrument $\texttt{brthord}_i$ must be correlated with $\texttt{educ}_i$ after we control for the exogenous covariates, which in our case is just $\texttt{sibs}_i$. Note that if we do not reject H_0, and $\pi_2 = 0$, then

$$\texttt{lwage}_i = (\beta_0 + \beta_1\hat{\pi}_0) + (\beta_2 + \beta_1\hat{\pi}_1)\,\texttt{sibs}_i + u_i = \gamma_0 + \gamma_1\texttt{sibs}_i + u_i.$$

By estimating this regression we will obtain two OLS estimates, $\hat{\gamma}_0$ and $\hat{\gamma}_1$. However, we have three unknown parameters we want to identify, $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\beta}_2$, and only two equations to solve:

$$\hat{\gamma}_0 = \hat{\beta}_0 + \hat{\beta}_1\hat{\pi}_0, \tag{41}$$

$$\hat{\gamma}_1 = \hat{\beta}_2 + \hat{\beta}_1\hat{\pi}_1. \tag{42}$$

Since a system of two equations with three unknowns does not have a unique solution, β_1 and β_2 cannot be identified separately.

(e) Estimate regression (34) using 2SLS, with $\texttt{brthord}_i$ as an instrument for $\texttt{educ}_i$ (and $\texttt{sibs}_i$ as its own instrument). Interpret your results.

Solution: The predicted values for education from the first stage is given by

$$\widehat{\texttt{educ}}_i = \hat{\pi}_0 + \hat{\pi}_1\texttt{sibs}_i + \hat{\pi}_2\texttt{brthord}_i. \tag{43}$$

and the corresponding second stage regression is

$$\texttt{lwage}_i = \beta_0 + \beta_1\widehat{\texttt{educ}}_i + \beta_2\texttt{sibs}_i + u_i.$$

The IV estimation output is presented in column (7) of Table 2, generated using

```
>> ivreg lwage sibs (educ = brthord)
```

The standard error on the returns to education coefficients, $\hat{\beta}_1 = 0.137$, is much larger than the one we had in part (c), and the corresponding test statistic is $t_{\hat{\beta}_1} = 1.83$. Also for $\hat{\beta}_2$, the standard error is very large relative to the point estimate, rendering $\texttt{sibs}_i$ strongly insignificant, since $t_{\hat{\beta}_2} = 0.12$ with a p-value of 0.067.

(f) Using the fitted values of $\widehat{\texttt{educ}}_i$ from the first-stage regression in part (d), compute the correlations between $\widehat{\texttt{educ}}_i$, $\texttt{sibs}_i$ and $\texttt{brthord}_i$. Use this result to explain your findings from part (h).

Solution: Recall that the predicted values of education from the first-stage regression are given by

$$\widehat{\texttt{educ}}_i = \hat{\pi}_0 + \hat{\pi}_1\texttt{sibs}_i + \hat{\pi}_2\texttt{brthord}_i.$$

Then we can run the following code to compute the

```
>> reg educ sibs brthord
>> predict educhat
>> corr educhat sibs brthord
```

Near multicollinearity causes large standard errors, and in the limit (in the case of perfect collinearity) we cannot estimate the coefficients of the model. Here $\widehat{\texttt{educ}}_i$ is highly correlated with $\texttt{sibs}_i$ because it is the projection of education on $\texttt{brthord}_i$, which is closely related to $\texttt{sibs}_i$.

Table 3: Correlation matrix

correlation	educhat	sibs	brthord
educhat	1.0000		
sibs	−0.9295	1.0000	
brthord	−0.8488	0.5939	1.0000

S4.4 Another bias in the wall: we don't need no education...without ability. The following log wage equation explicitly recognizes that ability ($\texttt{abty}_i$) affects log wages ($\texttt{lwage}_i$)

$$\texttt{lwage}_i = \beta_0 + \beta_1 \texttt{educ}_i + \beta_2 \texttt{abty}_i + u_i, \quad i = 1, \ldots, n. \tag{44}$$

In this model we hold ability fixed when measuring returns to education ($\texttt{educ}_i$).

(a) Assuming that the primary interest is in obtaining a reliable estimate of the slope parameters β_1, and that there is no direct measurement for ability, explain how you would do this using a method based on (1) a proxy variable; (2) an IV estimator. In doing so evaluate the following statement: "whilst IQ is a good candidate as a proxy variable for ability, it is not a good instrumental variable for education".

(b) Suppose now that the observations for IQ are not available to be used as either a proxy or an instrumental variable for ability and the following regression is estimated where ability is omitted:

$$\texttt{lwage}_i = \gamma_0 + \gamma_1 \texttt{educ}_i + v_i, \quad i = 1, \ldots, n. \tag{45}$$

Suppose the assumptions of the Gauss-Markov theorem are satisfied for (44). Show that the omission of $\texttt{abty}_i$ from the true regression (44) leads to endogeneity of $\texttt{educ}_i$ in the estimated regression (45).

Solution: Generally, an omitted variable leads to endogeneity in the estimated regression. It is a pathology of a linear regression model where the error term is not independent from the explanatory variable. In our case, the error from the estimated regression (45), v_i, is correlated with the explanatory variable $\texttt{educ}_i$. To show this, first note that the residual in (45) can be expressed as $v_i = \beta_2 \texttt{abty}_i + u_i$. Then, compute $\text{Cov}(v_i, \texttt{educ}_i)$ using the fact that $\texttt{educ}_i$ is exogenous in the original regression and so $\text{Cov}(u_i, \texttt{educ}_i) = 0$:

$$\text{Cov}(v_i, \texttt{educ}_i) = \text{Cov}(\beta_2 \texttt{abty}_i + u_i, \texttt{educ}_i).$$

$$
\begin{aligned}
&= \beta_2 \mathrm{Cov}\left(\texttt{abty}_i, \texttt{educ}_i\right) + \mathrm{Cov}\left(u_i, \texttt{educ}_i\right) \\
&= \beta_2 \mathrm{Cov}\left(\texttt{abty}_i, \texttt{educ}_i\right) \neq 0
\end{aligned}
$$

This term is non-zero since education is correlated with ability. Hence, the assumption of exogeneity of $\texttt{educ}_i$ in regression (45) is violated. This causes OLS estimators to be biased and inconsistent.

(c) In this question you will show that, in general, the OLS estimator $\hat{\gamma}_1$ from (45) is biased for β_1 and derive the magnitude of the bias. For simplicity of exposition, denote $y_i = \texttt{lwage}_i$, $x_{1i} = \texttt{educ}_i$ and $x_{2i} = \texttt{abty}_i$, and assume that the explanatory variables are fixed numbers. Then start by considering the OLS estimator $\hat{\gamma}_1$ from (45) given by:

$$
\hat{\gamma}_1 = \frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) y_i}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}. \tag{46}
$$

Solution: The expressions in this derivation will be quite long, so for simplicity of notation, use the hint. In this new notation the true model and the estimated regression can be expressed as

$$
\text{True model:} \quad y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i, \tag{47}
$$

$$
\text{Estimated model:} \quad y_i = \gamma_0 + \gamma_1 x_{1i} + v_i. \tag{48}
$$

The residuals from the two regressions are related as $v_i = \beta_2 x_{2i} + u_i$. Also, note that since the explanatory variables are fixed numbers, the assumption of exogeneity simply means that $\mathbb{E}\left[u_i\right] = 0$, for all i. Consider the OLS estimator of the slope coefficient from (48):

$$
\begin{aligned}
\hat{\gamma}_1 &= \frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) y_i}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2} \\
&= \frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) \left(\beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i\right)}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2} \\
&= \beta_0 \left[\frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}\right] + \beta_1 \left[\frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) x_{1i}}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}\right] + \beta_2 \left[\frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) x_{2i}}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}\right] \\
&\quad + \frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) u_i}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}.
\end{aligned}
$$

This can be simplified. The first term is zero, as $\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) = 0$. The second term is β_1, as $\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) x_{1i} = \sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2$. Hence,

$$
\hat{\gamma}_1 = \beta_1 + \beta_2 \left[\frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) x_{2i}}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}\right] + \frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) u_i}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}. \tag{49}
$$

Note that the second term can be expressed as

$$
\beta_2 \left[\frac{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) x_{2i}}{\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}\right] = \beta_2 \left[\frac{\frac{1}{n}\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right) x_{2i}}{\frac{1}{n}\sum_{i=1}^n \left(x_{1i} - \bar{x}_1\right)^2}\right] = \beta_2 \left[\frac{S_{x_1 x_2}}{S_{x_1 x_1}}\right].
$$

Observe that the term in the brackets, $S_{x_1x_2}/S_{x_1x_1}$, is the OLS estimator of the slope coefficient δ_1 from the regression of x_2 (ability) on x_1 (education):

$$x_{2i} = \delta_0 + \delta_1 x_{1i} + \varepsilon_i, \quad i = 1, \ldots, n. \tag{50}$$

Assume that x_{1i} is exogenous with respect to ε_i, $\mathbb{E}[\varepsilon_i|\mathbf{x}_1] = 0$, for all i. This implies that the OLS estimator, $\hat{\delta}_1 = S_{x_1x_2}/S_{x_1x_1}$, is unbiased for δ_1, namely $\mathbb{E}[\hat{\delta}_1] = \delta_1$. Now, returning to $\hat{\gamma}_1$ in expression (49), which can be simplified as

$$\hat{\gamma}_1 = \beta_1 + \beta_2\hat{\delta}_1 + \frac{\sum_{i=1}^n (x_{1i} - \bar{x}_1)\, u_i}{\sum_{i=1}^n (x_{1i} - \bar{x}_1)^2}. \tag{51}$$

In order to find the bias, take the expectation on both sides of this equation and recall that we made the assumption that the explanatory variables are fixed numbers, so they can be taken outside of the expectation operator:

$$\begin{aligned}
\mathbb{E}[\hat{\gamma}_1] &= \mathbb{E}\left[\beta_1 + \beta_2\hat{\delta}_1 + \frac{\sum_{i=1}^n (x_{1i} - \bar{x}_1)\, u_i}{\sum_{i=1}^n (x_{1i} - \bar{x}_1)^2}\right] \\
&= \mathbb{E}[\beta_1] + \mathbb{E}\left[\beta_2\hat{\delta}_1\right] + \mathbb{E}\left[\frac{\sum_{i=1}^n (x_{1i} - \bar{x}_1)\, u_i}{\sum_{i=1}^n (x_{1i} - \bar{x}_1)^2}\right] \\
&= \beta_1 + \beta_2\mathbb{E}\left[\hat{\delta}_1\right] + \left(\frac{1}{\sum_{i=1}^n (x_{1i} - \bar{x}_1)^2}\right)\mathbb{E}\left[\sum_{i=1}^n (x_{1i} - \bar{x}_1)\, u_i\right] \\
&= \beta_1 + \beta_2\mathbb{E}\left[\hat{\delta}_1\right] + \left(\frac{1}{\sum_{i=1}^n (x_{1i} - \bar{x}_1)^2}\right)\sum_{i=1}^n (x_{1i} - \bar{x}_1)\,\mathbb{E}[u_i] \\
&= \beta_1 + \beta_2\mathbb{E}\left[\hat{\delta}_1\right].
\end{aligned}$$

To simplify the third term, we used the assumption of exogeneity of education in the original regression (44), which implies that $(x_{1i} - \bar{x}_1)\,\mathbb{E}[u_i] = 0$ for all i. Finally, we know that the OLS estimators in (50) is unbiased if the assumption of $\mathbb{E}[\varepsilon_i|x_{1i}] = 0$ in regression (50) holds, and so $\mathbb{E}[\hat{\delta}_1] = \delta_1$. The omitted variable bias of the estimator $\hat{\gamma}_1$ for the population coefficient β_1 is then given by $\beta_2\delta_1$, since

$$\mathbb{E}[\hat{\gamma}_1] = \beta_1 + \beta_2\delta_1. \tag{52}$$

(d) Use economic reasoning to justify whether the omitted variable bias is likely to be positive or negative.

Solution: Since ability has a positive effect on wage ($\beta_2 > 0$), and education is positively correlated with ability ($\delta_1 > 0$), the bias will be positive ($\beta_2\delta_1 > 0$). We conclude that the coefficient from the regression where ability is omitted will overestimate the true effect of education on log wages, ($\hat{\gamma}_1 > \beta_1$).

(e) Under what conditions would $\hat{\gamma}_1$ yield an unbiased estimator for β_1? Are these conditions realistic?

Solution: From (52), observe that $\hat{\gamma}_1$ is an unbiased estimator for β_1 only if the bias is zero, $\beta_2\delta_1 = 0$. This can happen in two cases. First, if $\beta_2 = 0$ and $\texttt{abty}_i$ has no

effect on $\texttt{lwage}_i$, which means that ability was correctly omitted from the estimated regression (45). This is false by construction, as we assumed that the data generating process in (44) contains $\texttt{abty}_i$. Second, if $\delta_1 = 0$, which means $\texttt{educ}_i$ and $\texttt{abty}_i$ are uncorrelated, $\text{Cov}\left(\texttt{educ}_i, \texttt{abty}_i\right) = 0$. This is probably not true either, unless labour economists have been lying to us all this time. Therefore, in general, we would expect $\hat{\gamma}_1$ to yield a biased estimator of the partial effect of education on log wages with the expected bias of size $\beta_2\delta_1$.

S4.5 "And the award for the Best Test for Heteroskedasticity goes to..."
Consider the following linear model with a single explanatory variable X_i:

$$y_i = \alpha + \beta x_i + u_i, \quad i = 1, \ldots, n. \tag{53}$$

(a) Describe two different procedures which you could adopt to test whether the OLS residuals from this regression are heteroskedastic. Discuss the relative advantages or disadvantages of your proposed tests.

Solution: Two possible procedures are Breusch-Pagan test and White's test. In both cases, we first estimate α, β from the original regression in (53) using OLS. We then obtain the predicted values $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$, and the corresponding fitted residuals $\hat{u}_i = y_i - \hat{y}_i$. First, note that by construction of the OLS methods, the mean of the fitted residuals is zero, $\bar{\hat{u}} = 0$. Hence, the variance of u_i is the same thing as the mean of the squares of u_i. To test for heteroskedasticity, we conjecture that the variance of the error term is not constant, but is some function of the explanatory variable, $u_i^2 = f\left(x_i\right)$. The Breush-Pagan and the White's tests differ in the treatment of the specification of the function $f\left(\cdot\right)$. In the Breusch-Pagan test, $f\left(\cdot\right)$ is linear in x_i, and in the White's test it is non-linear. Table (4) provides details of the estimated auxiliary regressions, null and alternative hypotheses, test statistics and rejection rules for both tests. Note that this table is only applicable for the case with $k = 1$ explanatory variables. A more general case for the Breush-Pagan test for k explanatory variables

Table 4: Breusch-Pagan and White's tests for heteroskedastic errors

	Breusch-Pagan test	**White's test**
Auxiliary model:	$\hat{u}_i^2 = \gamma_0 + \gamma_1 x_i + \cdots + \gamma_k x_{ki} + \varepsilon_i$	$\hat{u}_i^2 = \delta_0 + \delta_1 \hat{y}_i + \delta_2 \hat{y}_i^2 + \varepsilon_i$
$\text{Var}\left(u_i\right) = \sigma^2$:	$H_0\colon\ (\gamma_1 = 0) \cap \cdots \cap (\gamma_k = 0)$	$H_0\colon\ (\delta_1 = 0) \cap (\delta_2 = 0)$
$\text{Var}\left(u_i\right) \neq \sigma^2$:	$H_1\colon\ (\gamma_1 \neq 0) \cup \cdots \cup (\gamma_k \neq 0)$	$H_1\colon\ (\delta_1 \neq 0) \cup (\delta_2 \neq 0)$
F-test statistic:	$V_1 = \frac{R^2/k}{(1-R^2)/(n-k-1)} \sim F_{k,n-k-1}$	$V_1 = \frac{R^2/2}{(1-R^2)/(n-3)} \sim F_{2,n-3}$
Reject H_0 at α:	$v_1 > F_{1-\alpha,k,n-k-1}$	$v_1 > F_{1-\alpha,2,n-3}$
χ^2-test statistic:	$V_2 = nR^2 \sim \chi_k^2$	$V_2 = nR^2 \sim \chi_2^2$
Reject H_0 at α:	$v_2 > \chi^2_{1-\alpha,k}$	$v_2 > \chi^2_{1-\alpha,2}$

The null hypothesis corresponds to homoskedastic errors, their variance is constant across all i. The alternative corresponds to heteroskedastic errors, their variance is not constant and depends on the explanatory variable. The Breusch-Pagan tests for the significance of the linear effects of the independent variables on squared fitted residuals, i.e. testing whether the variance of the residuals is a linear function of the exogenous variable. An alternative to the standard t-test, one could perform an F-test with one restriction and the corresponding limiting distribution $F_{1,n-1}$. Note, that this is the only possibility if we have more than one regressor in (53), for example if $\hat{u}_i^2 = \gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{2i} + \varepsilon_i$ and test H_0:

White tests for significance of non-linear effects of the exogenous variables on squared residuals. An alternative to an F-test with two restrictions presented in the table, one could use a Lagrange-multiplier (LM) test, with the test statistic $LM = nR^2 \sim \chi_2^2$, where R^2 is from the auxiliary regression and χ_2^2 is the limiting distribution. White's test is more general, and cleverly uses the predicted dependent variables, which means the auxiliary regression can be written as a polynomial in the exogenous variables:

$$\hat{u}_i^2 = \delta_0 + \delta_1 \hat{y}_i + \delta_2 \hat{y}_i^2 + v_i = \delta_0 + \delta_1 \left(\hat{\alpha} + \hat{\beta} x_i\right) + \delta_2 \left(\hat{\alpha} + \hat{\beta} x_i\right)^2 + v_i \tag{54}$$

which can be simplified in a reduced form as

$$\hat{u}_i^2 = \pi_0 + \pi_1 x_i + \pi_2 x_i^2 + v_i, \tag{55}$$

where π_j, $j = 0, 1, 2$, are combinations of the original coefficients $\delta_0, \delta_1, \delta_2, \alpha, \beta$. In this specification, we allow residual variance to depend on x_i^2 (compared to just x_i in the Breusch-Pagan test). If the original regression (53) includes more than one exogenous variable, say x_1 and x_2, then the auxiliary regression will also implicitly include x_{2i}^2 as well as the cross-term $x_{1i}x_{2i}$. The White's test can also include higher polynomials in $\hat{y}_i$, such as for example $\hat{y}_i^3$, in which case the null hypothesis contains three restrictions, and the limiting distribution of the test statistic becomes $F_{3,n-4}$. White's test loses power quickly when the number of exogenous variables goes up.

(b) Outline how you would modify the specification of your model, or the estimation procedure, if your test showed significant evidence of heteroskedasticity.

Solution: Textbook work: use robust errors if the functional form of heteroskedasticity is unknown or Generalized Least Squares if it is known (which is never the case in real empirical applications).

(c) A researcher, who has access to a dataset containing observations on log labour demand ($\texttt{llab}_i$), log capital ($\texttt{lcap}_i$), log output ($\texttt{lout}_i$) and average log wages ($\texttt{lwage}_i$) for a sample of 569 firms, computes the following regressions using OLS:

$$\begin{aligned} \texttt{llab}_i &= \underset{(0.071)}{6.177} + \underset{(0.026)}{0.990}\texttt{lout}_i - \underset{(0.019)}{0.004}\texttt{lcap}_i - \underset{(0.071)}{0.928}\texttt{lwage}_i + \hat{u}_i, \\ SSR &= 122.24, \qquad SST = 779.09, \qquad R^2 = 0.8430 \end{aligned} \tag{56}$$

where $\hat{u}_i$ are the OLS estimated residuals. Next, an auxiliary regression of $\hat{u}_i^2$ on the three original explanatory variables from (56), their squares and interaction terms is estimated and the following estimation results are obtained: $SSR = 404.53$, $SST =$

450.93 and $R^2 = 0.1029$. Use this information to conduct a statistical test (at the 5% significance level) of the null hypothesis that the disturbances in regression (56) are homoskedastic. Explain the reasoning behind your choice of the test statistic.

Solution: This is a version of White's heteroskedasticity test, the explanatory variables, their squares and interaction terms are included explicitly in the auxiliary regression for the square of the errors. The following auxiliary regression is estimated using OLS

$$\hat{u}_i^2 = \gamma_0 + \mathbf{Z}_i^\intercal \boldsymbol{\gamma} + \varepsilon_i, \quad i = 1, \ldots, n, \tag{57}$$

where ε_i are the error terms, $\mathbf{Z}_i$ is a 9×1 vector with explanatory variables, their squares and interaction, $\boldsymbol{\pi}$ is a 9×1 vector of parameters:

$$\mathbf{Z}_i = \begin{bmatrix} \texttt{lout}_i \\ \texttt{lcap}_i \\ \texttt{lwage}_i \\ \texttt{lout}_i^2 \\ \texttt{lcap}_i^2 \\ \texttt{lwage}_i^2 \\ \texttt{lout}_i \times \texttt{lcap}_i \\ \texttt{lout}_i \times \texttt{lwage}_i \\ \texttt{lcap}_i \times \texttt{lwage}_i \end{bmatrix}, \qquad \boldsymbol{\gamma} = \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \\ \gamma_4 \\ \gamma_5 \\ \gamma_6 \\ \gamma_7 \\ \gamma_8 \\ \gamma_9 \end{bmatrix}.$$

The joint testing is then conducted on the following hypothesis:

$$\begin{aligned} H_0 &: \ (\gamma_1 = 0) \cap \cdots \cap (\gamma_k = 0) \implies \text{homoskedasticity} \\ H_1 &: \ (\gamma_1 \neq 0) \cup \cdots \cup (\gamma_k \neq 0) \implies \text{heteroskedasticity} \end{aligned}$$

Under the null of homoskedasticity, the regressors have no explanatory power for the variance of the error terms. Hence, the restricted regression that contains only the intercept (as all the other coefficients have been set to zero), $\hat{u}_i = \gamma_0 + \varepsilon_i$, will have $R_r^2 = 0$. The first way to do the test is using the standard F-test statistic. The second way is using the χ^2-test. Both statistics use the R^2 from the unrestricted (auxiliary) regression for the squares of fitted residuals in (57) and the number of explanatory variables $k = 9$, and the sample size $n = 569$. These are given by

$$V_1 = \frac{R^2/k}{(1 - R^2)\,/\,(n - k - 1)} \sim F_{k,n-k-1}, \tag{58}$$

$$V_2 = nR^2 \sim \chi_k^2. \tag{59}$$

For the significance level $\alpha = 0.0$, the first test rejects the null if the observed sample test statistic $v_1 > F_{1-\alpha,k,n-k-1}$. The second test reject when $v_2 > \chi^2_{1-\alpha,k}$. Compute the observed test statistics

$$\begin{aligned} v_1 &= \frac{0.1029/9}{(1 - 0.1029)\,/559} = 7.1243 > 1.8966 = F_{0.95,9,559}, \\ v_2 &= 569 \times 0.1029 = 58.5501 > 16.919 = \chi^2_{0.95,9}. \end{aligned}$$

In both cases the tests statistic exceeds the $(1 - \alpha)$ critical value and so the null of homoskedasticity is rejected at $\alpha = 0.05$. We conclude that the variance of the error terms depends on the explanatory variables, or in other words the errors are heteroskedastic.

(d) What are the implications of the result which you obtained in part (c) for the properties of the OLS estimators in (56) and their associated standard errors?

Solution: By now we know that heteroskedasticity does not bias the OLS estimators, but does bias the corresponding standard errors and hence invalidates the standard inference on the regression coefficients.

S4.6 Does size matter? An empirical investigation.

An investigator has access to cross-section data on chief executive remuneration for a sample of 209 US companies in 1990. He estimates the following regression:

$$\texttt{lsal}_i = \beta_0 + \beta_1 \texttt{lsales}_i + \beta_2 \texttt{roe}_i + \beta_3 \texttt{pcroe}_i + \beta_4 \texttt{util}_i + \beta_5 \texttt{fin}_i + \beta_6 \texttt{ind}_i + \varepsilon_i, \quad (60)$$

where $\varepsilon_i \sim \text{iid}\, N\left(0, \sigma^2\right)$ and the variables are defined as follows:

`lsal`	log of salary for ith the chief executive in the sample
`lsales`	log of company sales
`roe`	return on equity
`pcroe`	percentage change in roe 1989-1990
`util`	dummy: equals 1 if firm is a utility, zero otherwise
`fin`	dummy: equals 1 if firm is in finance, zero otherwise
`ind`	dummy: equals 1 if firm produces industrial products, zero otherwise

The data are sorted by firm size. Firms 1-128 are small firms with sales less than \$5 billion. Firms 198-209 are large firms with sales in excess of \$20 billion. The remaining firms (129-197) are classified as medium. Table 5 gives summary statistics for a number of regressions conducted by the investigator: SSR denotes the residual sum of squares.

Table 5: OLS estimates of the regression for $\texttt{lsalary}_i$

	A	B	C	D
Sample	1-209	1-128	129-209	1-197
R^2	0.3632	0.3542	0.1485	0.3577
SSR	42.487	24.063	15.819	39.715

(a) Use Table 5 to test the hypothesis that the parameters of the models do not differ significantly between small firms and the combined set of medium and large firms.

Solution: Conduct a Chow test, where model A is the restricted model or the combined data with SSR_A, while model B and model C represent the two subgroups, or the unrestricted model, with the corresponding SSR_B and SSR_C, $n_B = 128$ and $n_C = 81$. Regression (60) is estimated for the three subsamples:

$$\texttt{lsal}_i = \beta_0^j + \beta_1^j \texttt{lsales}_i + \beta_2^j \texttt{roe}_i + \beta_3^j \texttt{pcroe}_i + \beta_4^j \texttt{util}_i + \beta_5^j \texttt{fin}_i + \beta_6^j \texttt{ind}_i + \varepsilon_i,$$

where j can be either A, B or C. We want to test the hypothesis that the population parameters for the small firms are the same as those for medium/large firms, formally:

$$H_0 : \left(\beta_0^B = \beta_0^C\right) \cap \left(\beta_1^B = \beta_1^C\right) \cap \cdots \cap \left(\beta_6^B = \beta_6^C\right)$$

$$H_1 : \left(\beta_0^B \neq \beta_0^C\right) \cup \left(\beta_1^B \neq \beta_1^C\right) \cup \cdots \cup \left(\beta_6^B \neq \beta_6^C\right)$$

so a total of $q = k + 1 = 7$ restrictions that the coefficients in the two subsamples are equal. Use the following Chow-test statistic

$$V = \frac{\left[SSR_A - (SSR_B + SSR_C)\right] / (k+1)}{(SSR_B + SSR_C) / (n_B + n_C - 2(k+1))} \sim F_{k+1, n-2(k+1)} \tag{61}$$

The observed test statistic $v = 1.820 < F_{0.95,7,195} = 2.010$, for significance level $\alpha = 0.05$, and so we fail to reject H_0. However, the critical value for significance of $\alpha = 0.1$, given by $F_{0.9,7,195} = 1.748$, is smaller than the computed test statistic and so with with significance level the null is rejected. We do have some evidence, albeit weak, that the coefficients in the two subsamples are not the same.

(b) Use Table 5 to test the hypothesis that a model estimated over a sample of small and medium size firms can also be used to predict the behavior of large firms.

Solution: This is about the predictive failure test, which test for the stability of parameters between the full sample and a long sub-sample. Unlike the Chow test, this requires the estimation of the full sample and only one of the sub-samples. We want to test whether the population parameters for small/medium firms are the same as those for large firms. Using the information presented in the Table, we can do that by testing Model A (full sample / restricted): the parameters for all three types of firms are restricted to be the same, against Model D (sub-sample / unrestricted): where the parameters of the small/medium firms are allowed to differ from those of the large firms. Formally,

$$H_0 : \left(\beta_0^A = \beta_0^D\right) \cap \left(\beta_1^A = \beta_1^D\right) \cap \cdots \cap \left(\beta_6^A = \beta_6^D\right)$$
$$H_1 : \left(\beta_0^A \neq \beta_0^D\right) \cup \left(\beta_1^A \neq \beta_1^D\right) \cup \cdots \cup \left(\beta_6^A \neq \beta_6^D\right)$$

Intuitively, the null hypothesis says that if we used Model D (small/medium firms) to forecast salaries for the large firms, all the prediction errors for all the foretasted observations should be zero. Conduct an F-test with the test statistic

$$V = \frac{(SSR_A - SSR_D) / (n_A - n_D)}{SSR_D / (n_D - k - 1)} \sim F_{n_A - n_D, n_D - k - 1} \tag{62}$$

where $n_A = 209$, $n_D = 197$ and $k = 6$. Note that in the numerator, the number of degrees of freedom is $(n_A - n_D)$ and not $k+1$ as we would except in the usual F-test. This is because the number of observations in the restricted and the unrestricted models differ by the number of observations that must be excluded from the full sample in order to arrive to the size of the sub-sample. In other words, this is the number of large firms, or the number of observations in the full sample that needs to be forecasted from the sub-sample model. Doing the computations, $v = 1.105 < F_{0.95,24,190} = 1.575$, and again do not reject the null hypothesis that the parameters of the models are equal at 5% significance. It seems that the salaries in all types of firms are described by the same linear population model.

(c) Describe how you could extend the investigators model to test the hypothesis that the effect of log sales on log salary differs between small and large firms.

Solution: If we were allowed to change the model specification slightly, we could construct a dummy variable for small firms:

$$\mathtt{D}_i^s = \begin{cases} 1 & \text{small firm} \\ 0 & \text{otherwise} \end{cases}$$

Add this dummy as an interaction with $\mathtt{lsales}_i$ in regression (60) and estimate the following specification on the sample that only includes small and large firms:

$$\mathtt{lsal}_i = \beta_0 + \beta_1 \mathtt{lsales}_i + \cdots + \beta_6 \mathtt{ind}_i + \delta_s \left(\mathtt{D}_i^s \times \mathtt{lsales}_i\right) + \varepsilon_i. \tag{63}$$

Finally, test $H_0 \colon \delta_s = 0$ against $H_1 \colon \delta_s \neq 0$ using a t-test.

S4.7 Dummy Wars. Episode I: The large-class menace.

A researcher wants to analyze the determinants of test scores in primary schools. He collects a dataset for 420 California school districts. Table 6 describes the dataset, it contains data on district-average reading and mathematics tests performance for 5th grade students, school characteristics, student demographic backgrounds and district economic position. OLS is used to estimate a series of regressions, starting with model (1) in Table 7, which reports the estimated coefficients from a regression of district average reading scores, `read`, on school characteristics, student demographic backgrounds and log of district average income (standard errors are given in parenthesis).

Table 6: Variables description and summary statistics

Var	Description	Mean	St.Dev.
`read`	average score on a standardized reading test	655.0	20.1
`math`	average score on a standardized maths test	653.3	18.8
`str`	student-teacher ratio	19.6	1.9
`esl`	% students for whom English is a 2nd lang.	15.8	18.3
`meal`	% students qualifying for free meals	44.7	27.1
`soc`	% students in households receiving welfare	13.2	11.5
`inc`	district average income ($1000's).	15.3	7.2
`linc`	ln (`inc`)		

(a) Compute and interpret the 95% confidence interval for the coefficient on `meal`. What does this interval suggest about statistical significance of $\beta_{\mathtt{meal}}$?

Solution: Since the sample size is large we can use the central limit theorem to derive the asymptotic distribution of the OLS estimators $\hat{\beta}$ for the population coefficient β,

$$\hat{\beta} \overset{a}{\sim} N\left(\beta, S_{\hat{\beta}}^2\right) \tag{64}$$

where $S_{\hat{\beta}}^2$ is the estimator of the variance of $\hat{\beta}$ with the corresponding (observed) standard error $s_{\hat{\beta}}$. Then, the $(1-\alpha)\,100\%$-confidence interval for $\beta_{\mathtt{meal}}$ is given by

$$\mathrm{CI}_{1-\alpha}\left(\beta_{\mathtt{meal}}\right) = \left[\hat{\beta}_{\mathtt{meal}} - z_{1-\frac{\alpha}{2}} \times s_{\hat{\beta}_{\mathtt{meal}}}, \hat{\beta}_{\mathtt{meal}} + z_{1-\frac{\alpha}{2}} \times s_{\hat{\beta}_{\mathtt{meal}}}\right], \tag{65}$$

where $\alpha = 0.05$ is the significance level, and we use the critical values of the standard normal distribution $z_{0.975} = 1.96$. The confidence interval is then given by

$$\mathrm{CI}_{0.95}\left(\beta_{\texttt{meal}}\right) = [-0.3937 \pm 1.96 \times 0.0378] = [-0.4677, -0.3197]\,.$$

This interval does not contain 0 and hence $\beta_{\texttt{meal}}$ is significant at the 5% level.

Table 7: OLS estimation results

	(1) read	(2) math	(3) read
const	665.1621	651.7782	649.7702
	(7.4847)	(9.0253)	(5.7282)
linc	11.4227	11.9584	11.9584
	(1.6979)	(2.0474)	(1.6979)
esl	−0.2598	−0.1167	−0.2578
	(0.0326)	(0.0393)	(0.0325)
meal	−0.3937	−0.3446	−0.3952
	(0.0378)	(0.0378)	(0.0377)
soc	−0.0269	−0.1138	−0.0278
	(0.0566)	(0.0455)	(0.0565)
str	−0.9343	−0.5759	
	(0.2253)	(0.2717)	
str_2			−0.9910
			(1.4092)
str_3			−4.5034
			(1.3867)
str_4			−5.3919
			(1.8470)
n	420	420	420
SSE	140081.25	104720.15	140425.53
SSR	29333.38	42650.57	28989.10
SST	169414.63	147370.72	169414.63
R^2	0.8269	0.7106	0.8289
R^2-adj.	0.8248	0.7071	0.8269

standard errors in parentheses

(b) Test at the 10% significance level the hypothesis that all other things being equal, the percentage of children in the school district for whom English is a second language has no effect of the district average reading score.

Solution: We want to test significance of $\beta_{\texttt{esl}}$ at significance level $\alpha = 0.1$:

$$H_0\colon \beta_{\texttt{esl}} = \beta^0_{\texttt{esl}} = 0$$
$$H_1\colon \beta_{\texttt{esl}} \neq 0$$

Under the null, provided n is large enough, the central limit theorem suggests that the OLS estimate is approximately normal, and so the test-statistic is

$$T_{\hat{\beta}_{\texttt{esl}}} = \frac{\hat{\beta}_{\texttt{esl}} - \beta^0_{\texttt{esl}}}{S_{\hat{\beta}_{\texttt{esl}}}} \overset{a}{\sim} N(0,1). \tag{66}$$

Note that $\beta^0_{\texttt{esl}} = 0$ is the hypothesized population coefficient. Reject the null at $\alpha = 0.1$ if $|t_{\hat{\beta}_{\texttt{esl}}}| > z_{0.95} = 1.645$. The observed test statistic is

$$t_{\hat{\beta}_{\texttt{esl}}} = \frac{\hat{\beta}_{\texttt{esl}} - \beta^0_{\texttt{esl}}}{s_{\hat{\beta}_{\texttt{esl}}}} = \frac{-0.2598}{0.0326} = -7.964.$$

As $|t_{\hat{\beta}_{\texttt{esl}}}| = 7.964 > 1.645$, reject the null hypothesis at $\alpha = 0.1$ significance. We have sufficient evidence that the percentage of children in the school district for whom English is a second language has an effect of the district average reading score.

(c) Compute an estimate of the effect of a 10% increase in average district income on district average reading score.

Solution: Note that model (1) includes the log of income, $\texttt{linc} = \ln(\texttt{inc})$, hence:

$$\beta_{\texttt{linc}} = \frac{\partial \texttt{read}}{\partial \ln(\texttt{inc})} = \frac{\partial \texttt{read}}{\frac{\partial \texttt{inc}}{\texttt{inc}}} \implies \partial \texttt{read} = \beta_{\texttt{linc}} \left(\frac{\partial \texttt{inc}}{\texttt{inc}} \right).$$

The OLS estimate from model (1), $\hat{\beta}_{\texttt{linc}} = 11.4227$, and $\frac{\partial \texttt{inc}}{\texttt{inc}} = 0.1$ imply that $\partial \texttt{read} = 11.4227 \times 0.1 = 1.1423$.

(d) Predict the average reading score for a district in which the student-teacher ratio is 22 and all other school characteristics and student demographic variables take the sample mean value.

Solution: Use sample means from Table 6 and estimates from Table 7 to find:

$$\begin{aligned}\widehat{\texttt{read}} &= 665.1621 - 0.9343 \times 22 + 11.4227 \times \ln(15.3) - 0.2598 \times 15.8 \\ &\quad - 0.3937 \times 44.7\text{-}0.0269 \times 13.2 \\ \widehat{\texttt{read}} &= 653.7265.\end{aligned}$$

Note that mean `linc` is not available, so we have to approximate it with $\ln(\bar{\texttt{inc}})$, even thought the two are not exactly equal as log is a concave function.

(e) Model (2) in Table 7 reports the equivalent OLS regression results for district average mathematics score. Compare and contrast the OLS regression results obtained for the average reading scores with those for average mathematics scores. Conduct hypothesis testing that the coefficients in model (1) are statistically different from the corresponding coefficients in model (2), that is test $H_0\colon \beta_j^{\texttt{read}} = \beta_j^{\texttt{math}}$ against $H_1\colon \beta_j^{\texttt{read}} \neq \beta_j^{\texttt{math}}$, for $j = \texttt{linc}, \texttt{esl}, \texttt{meal}, \texttt{soc}, \texttt{str}$.

Solution: Observe that the student-teacher ratio appears to be more important for reading tests than maths tests. The effects of income and free meals look similar in both regressions. The magnitude of $\hat{\beta}_{\texttt{esl}}$ is about twice as great for reading, which makes sense since being a non-native speaker should be more of a challenge for reading

scores than for math. Being on welfare has a greater effect on maths scores. R^2 is higher for reading, perhaps due to reading performance being a more immediate consequence of the environment, while maths has a greater effect of some unobserved components that includes innate ability.

Formally, we should also test whether the coefficients in model (1) and model (2) are statistically different. We then test the following hypothesis: $H_0\colon \beta_j^{\texttt{read}} = \beta_j^{\texttt{math}}$ against $H_0\colon \beta_j^{\texttt{read}} \neq \beta_j^{\texttt{math}}$, using the t-statistics for the comparison of two sample means. In order to do that we must of course assume that the estimators of $\beta_j^{\texttt{read}}$ and $\beta_j^{\texttt{math}}$ are independent for each j, that is

$$\hat{\beta}_j^{\texttt{read}} \overset{a}{\sim} N\left(\beta_j^{\texttt{read}}, S^2_{\hat{\beta}_j^{\texttt{read}}}\right), \qquad \hat{\beta}_j^{\texttt{math}} \overset{a}{\sim} N\left(\beta_j^{\texttt{math}}, S^2_{\hat{\beta}_j^{\texttt{math}}}\right).$$

Then, we can formulate the following test statistic under the assumption that the null is true and hence that $\beta_j^{\texttt{read}} - \beta_j^{\texttt{math}} = 0$:

$$T_{\hat{\beta}_j} = \frac{\hat{\beta}_j^{\texttt{read}} - \hat{\beta}_j^{\texttt{math}}}{\sqrt{S^2_{\hat{\beta}_j^{\texttt{read}}} + S^2_{\hat{\beta}_j^{\texttt{math}}}}} \overset{a}{\sim} N(0,1), \tag{67}$$

Reject H_0 at $\alpha = 0.05$ if the observed test statistic is such that $|t_{\hat{\beta}_j}| > z_{0.95} = 1.96$. From Table 7 compute:

$$\begin{aligned}
\beta_{\texttt{linc}}: \quad t_{\hat{\beta}_{\texttt{linc}}} &= \frac{11.4227 - 11.9584}{\sqrt{1.6979^2 + 2.0474^2}} = -0.2014 && \Longrightarrow \text{Do not reject } H_0 \\
\beta_{\texttt{esl}}: \quad t_{\hat{\beta}_{\texttt{esl}}} &= \frac{-0.2598 - (-0.1167)}{\sqrt{0.0326^2 + 0.0393^2}} = -2.80 && \Longrightarrow \text{Reject } H_0 \\
\beta_{\texttt{meal}}: \quad t_{\hat{\beta}_{\texttt{meal}}} &= \frac{-0.3937 - (-0.3446)}{\sqrt{0.0378^2 + 0.0378^2}} = -0.9184 && \Longrightarrow \text{Do not reject } H_0 \\
\beta_{\texttt{soc}}: \quad t_{\hat{\beta}_{\texttt{soc}}} &= \frac{-0.0269 - (-0.1138)}{\sqrt{0.0566^2 + 0.0455^2}} = 1.1966 && \Longrightarrow \text{Do not reject } H_0 \\
\beta_{\texttt{str}}: \quad t_{\hat{\beta}_{\texttt{str}}} &= \frac{-0.9343 - (-0.5759)}{\sqrt{0.2253^2 + 0.2717^2}} = -1.0154 && \Longrightarrow \text{Do not reject } H_0
\end{aligned}$$

Conclude that there is only a significant difference in the way that the proportion of students for whom English is not a native language affects reading and maths tests scores, as we have postulated above.

(f) Model (3) in Table 7 reports the results for an alternative specification of the regression for the reading test scores, in which the continuous variable `srt` is replaced by a set of dummy variables:

$$\begin{aligned}
&\texttt{str_1} = 1 \text{ if } \texttt{str} < 17.5 \text{ and } 0 \text{ otherwise} \\
&\texttt{str_2} = 1 \text{ if } 17.5 \leq \texttt{str} < 19.5 \text{ and } 0 \text{ otherwise} \\
&\texttt{str_3} = 1 \text{ if } 19.5 \leq \texttt{str} < 2 \text{ and } 0 \text{ otherwise} \\
&\texttt{str_4} = 1 \text{ if } \texttt{str} \geq 22 \text{ and } 0 \text{ otherwise}
\end{aligned}$$

Interpret this regression and explain why the dummy variable `str_1` is not included in the regression equation.

Solution: Due to multicollinearity of the dummy variables we cannot include a full set of dummies and an intercept at the same time, and thus treat `str_1` as the default group. Holding other variables constant, the intercept in model (3) is the coefficient for the default/omitted group `str_1`, which is the situation when all the other dummies are zero. Note that the coefficients on `str_2`, `str_3`,and , `str_4` show the difference between the effect of `str_1` and the effect for the group in question. For example, $\hat{\beta}_{\texttt{str_2}} = -0.9910$ is the additional effect of being in a school district with an average student-teacher ratio between 17.5 and 19.5, as compared with a district with a student-teacher ratio lower than 17.5.

(g) Using model (3), compute an estimate of the effect on district average reading scores of increasing the student-teacher ratio from 16 to 18 and from 21 to 23.

Solution: The estimates in Table 7 suggest that a change from `str_1` to `str_2` leads to a change in the reading test score by $\hat{\beta}^{\texttt{read}}_{\texttt{str_2}} = -0.9910$. Similarly, a change from `str_3` to `str_4` leads to a change of

$$\hat{\beta}^{\texttt{read}}_{\texttt{str_4}} - \hat{\beta}^{\texttt{read}}_{\texttt{str_3}} = -5.3919 - (-4.5034) = -0.8885$$

(h) Overall, what do you infer from the results in model (1) and model (3) about the impact of an increase in the student-teacher ratio on average reading scores?

Solution: Model (1) assumes a linear effect of the student-teacher ratio on the reading test scores. The effect is negative and significant at the 1% level, since $t_{\hat{\beta}_{\texttt{str}}} = -4.147$. Model (3), on the other hand, relaxes that simplistic assumption and allows for the class-size effects to be non-linear. From the estimator results we notice some sort of threshold effects of the class size. To see which thresholds are important, conduct formal tests of significance in model (3). The coefficient on `str_2` is negative and not statistically significant; `str_3` is negative and statistically significant; `str_4` is negative and statistically significant at $\alpha = 0.01$, since

$$t_{\hat{\beta}_{\texttt{str_2}}} = -0.703, \quad t_{\hat{\beta}_{\texttt{str_3}}} = -3.248, \quad t_{\hat{\beta}_{\texttt{str_4}}} = -2.919.$$

Thus, the increase of the student-teacher ratio from below 17.5 to a range of $(17.5, 19.5)$ has no significant effect, while further increases to a class size of $(19.5, 22.0)$ or above 22.0 do. From these results, we can speculate that the threshold of significance is around 20 students per teacher. After this point the effect becomes significantly negative. The magnitude of the effect is initially very high for the jump from < 17.5 to $(19.5, 22)$, and then starts showing a diminishing marginal effect for a further increase in the class size.